

The empirical recurrence for OEIS A233719: recovering a conjecture that is not written down

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Abstract

OEIS A233719 records “Empirical recurrence of order 90”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 90, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 90 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233719 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having the sum of the squares of the edge differences equal to 26.”. It has offset 1 and begins

1820, 13588, 101644, 820524, 6622988, 54865696, 453172200, 3781440536, ...

The entry states:

Empirical recurrence of order 90 (see link above).

As of the “Last modified” line on the live entry (Jul 07 2014 03:12:06, revision 10) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 90 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	21	c_{24}	7412401028142935231	c_{47}	−235588326889269043869551907	c_{70}
c_2	106	c_{25}	−63200640584692062178	c_{48}	−877282571239778204139123254	c_{71}
c_3	−5095	c_{26}	−33996636061563789736	c_{49}	681752952752930569472736719	c_{72}
c_4	10121	c_{27}	594564984788388525960	c_{50}	1295316173913763346815121820	c_{73}
c_5	547211	c_{28}	−46803883268056748477	c_{51}	−1426948499748046468958297983	c_{74}
c_6	−2767718	c_{29}	−461277422243355153285	c_{52}	−1502881321252030347400394264	c_{75}
c_7	−33908027	c_{30}	2706470752589535728011	c_{53}	2319372329926821154110614250	c_{76}
c_8	261392661	c_{31}	29641641772732272198652	c_{54}	1276068763759777032558713768	c_{77}
c_9	1297920161	c_{32}	−30913835837000688926362	c_{55}	−2994272381678480268917071938	c_{78}
c_{10}	−14952220574	c_{33}	−157713607940837434806390	c_{56}	−614983139512806901528386927	c_{79}
c_{11}	−28138319704	c_{34}	235855819467662691391141	c_{57}	3081991041459832176825360695	c_{80}
c_{12}	593849250880	c_{35}	690562144197538623696716	c_{58}	−179123525447941814650354932	c_{81}
c_{13}	67125188551	c_{36}	−1382800459351058255642844	c_{59}	−2509368798312419598270680670	c_{82}
c_{14}	−17418469930323	c_{37}	−2449642364540797715387087	c_{60}	709656610608738521832457755	c_{83}
c_{15}	19374364158733	c_{38}	6527326586819830132897723	c_{61}	1583786965258269423730220623	c_{84}
c_{16}	390131457993501	c_{39}	6786725653658718933943803	c_{62}	−797669973801196817222539645	c_{85}
c_{17}	−822384572606082	c_{40}	−25312134787289597051856503	c_{63}	−743369000560797425618637613	c_{86}
c_{18}	−6790477234476028	c_{41}	−13251339152652853068395421	c_{64}	574970194549691705346286141	c_{87}
c_{19}	20776089268508448	c_{42}	81363795125024980320764454	c_{65}	235021539117137604180361293	c_{88}
c_{20}	92264057267414075	c_{43}	10431091687963756828990213	c_{66}	−293994385876590954123488499	c_{89}
c_{21}	−382669691490634417	c_{44}	−217351726423784445774772231	c_{67}	−33026980724542139507178871	c_{90}
c_{22}	−966204132726361299	c_{45}	43013093552582510674767516	c_{68}	108004039447996662206544015	
c_{23}	5487720863855475849	c_{46}	481398439394376484091224878	c_{69}	−9919927840129639835911656	

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 90, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $90 + 4$ terms removed returns order 90 as well, so $\deg q_{\min} = 90 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $90 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 90 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A233719.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.